Sed × precip suppression: 12 h runs (40 km to 1.25 km)

7 Nov 2026. Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz. Addendum to the 6 h technical note. Same 216-run matrix and forcing (the RF02 IOP file is constant over 0–12 h); means over hours 10–12. The first 6 h of each run reproduce the 6 h runs bit-for-bit (24 checked).

What changes at 12 h

1. **Steadiness.** At 2.5 and 1.25 km, $|d\mathrm{LWP}/dt| < 4\,\mathrm{g\,m^{-2}\,h^{-1}}$ over hours 10–12 in every run (Fig. **??**). These are close to equilibrium. At 40–10 km, LWP is still rising at high Nd with suppression on (up to $+18\,\mathrm{g\,m^{-2}\,h^{-1}}$; LWP reaches $340\,\mathrm{g\,m^{-2}}$).

2. **5 km has joined the resolved-cell regime.** By 12 h, LWP spatial SD is 70–140 $\mathrm{g\,m^{-2}}$, cloud cover drops at low Nd, and LWP and w_e look like 2.5 km. At 6 h, 5 km was still single-column-like. So the transition is resolution- *and* time-dependent.

3. **A falling branch at km scale.** With suppression on, $S_{200\to1000}$ is negative at both 2.5 and 1.25 km and grows with sedimentation: $-0.01/-0.03/-0.09$ at 1.25 km and $-0.01/-0.03/-0.06$ at 2.5 km for sed $\times 0.5/1/2$ (Fig. **??**). LWP peaks near Nd $\approx$ 200–500, while w_e keeps rising with Nd (to 6.9–7.2 $\mathrm{mm\,s^{-1}}$). At 6 h this branch was barely visible.

4. **The low-Nd branch steepens with sedimentation at km scale.** With suppression on, $S_{20\to50}$ is 0.57/0.65/0.88 at 1.25 km. With suppression off it is $-0.02/0.11/0.43$, all of it from sedimentation-driven deck break-up (cloud cover at Nd 20 with sed×2 falls below 0.6).

5. **Single-column limit (40–10 km).** LWP with suppression on rises strongly with Nd ($S_{100\to200} = 0.20$–0.35). With suppression off it is flat to slightly decreasing (S from -0.01 to -0.1). The sedimentation switch in w_e persists at a lower level ($\approx$4.0 vs. $\approx$3.0 $\mathrm{mm\,s^{-1}}$).

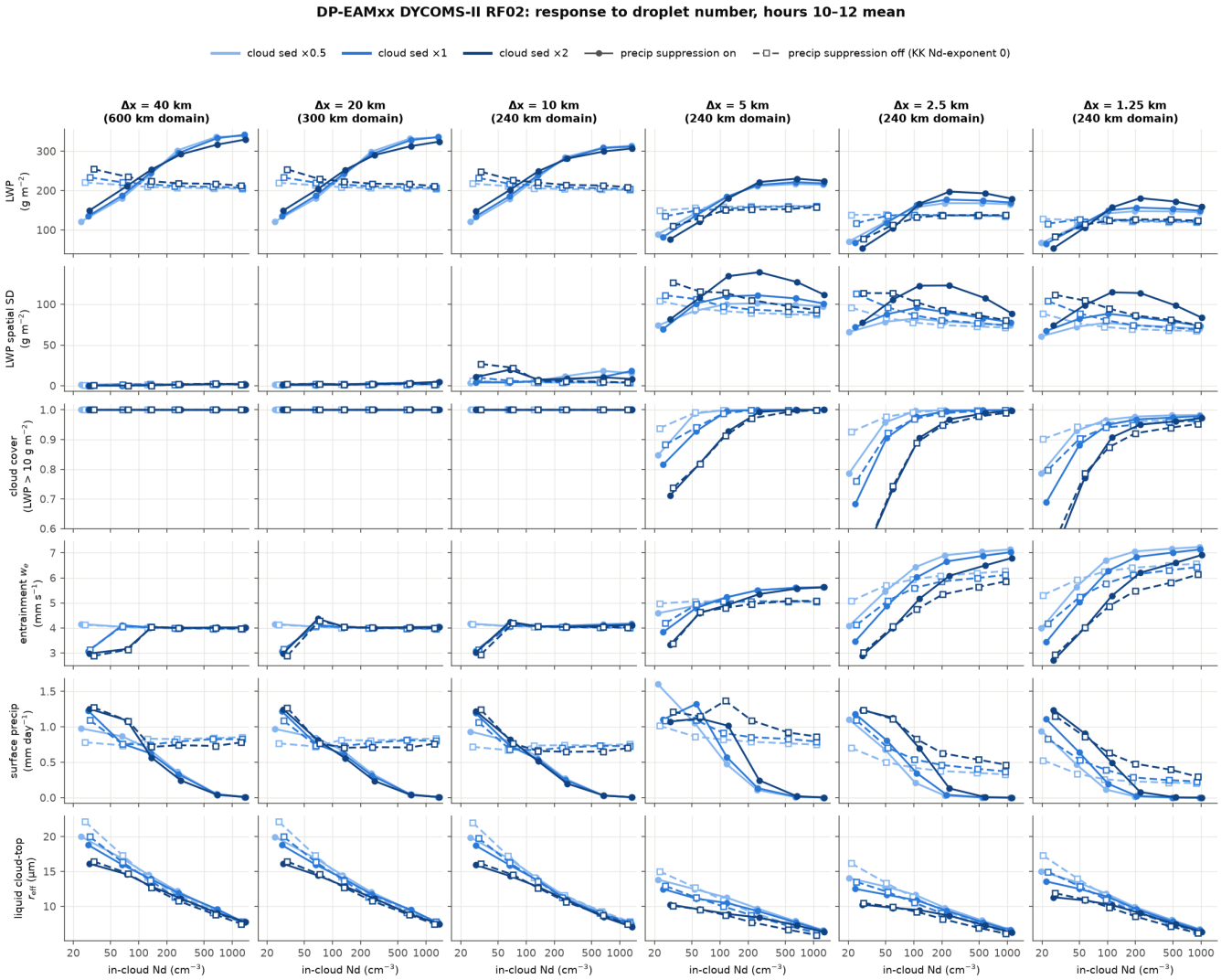


Figure 1: Hour 10–12 means against diagnosed in-cloud Nd. Same encoding as the 6 h note.

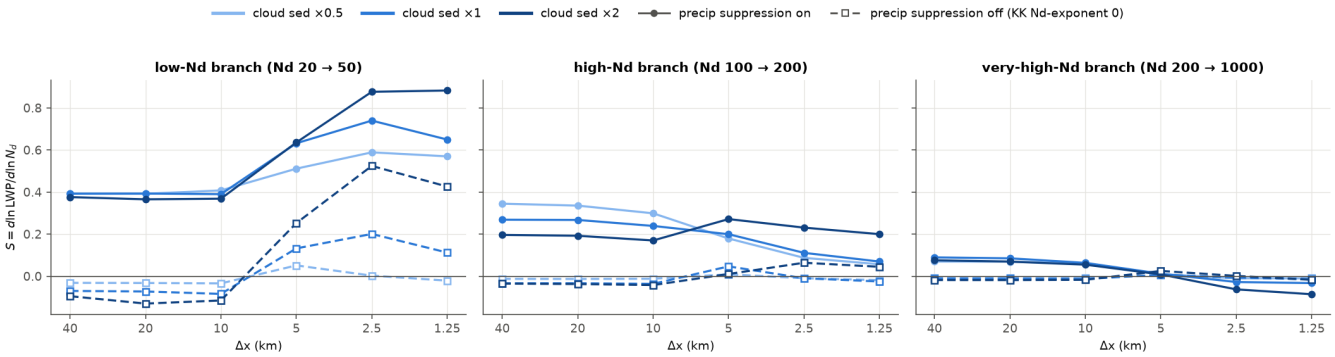


Figure 2: Susceptibility on three Nd branches against resolution, hours 10–12.

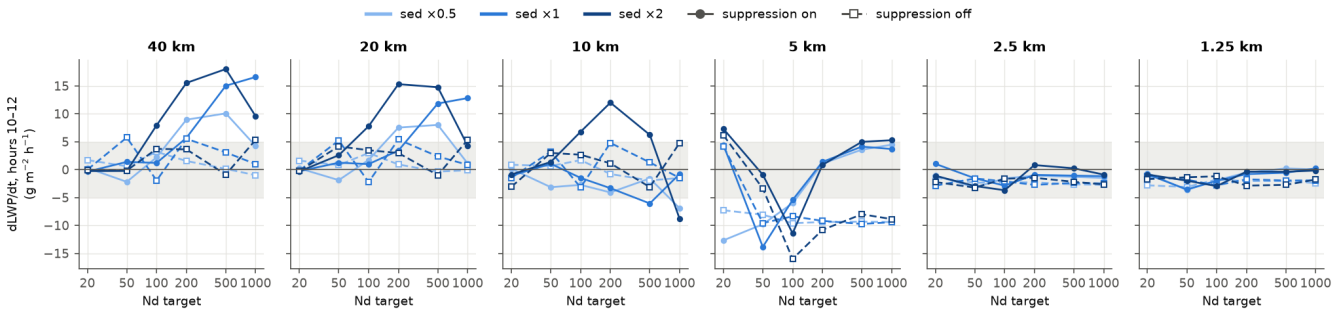


Figure 3: LWP trend over hours 10–12. The grey band marks $\pm 5\,\mathrm{g\,m^{-2}\,h^{-1}}$.

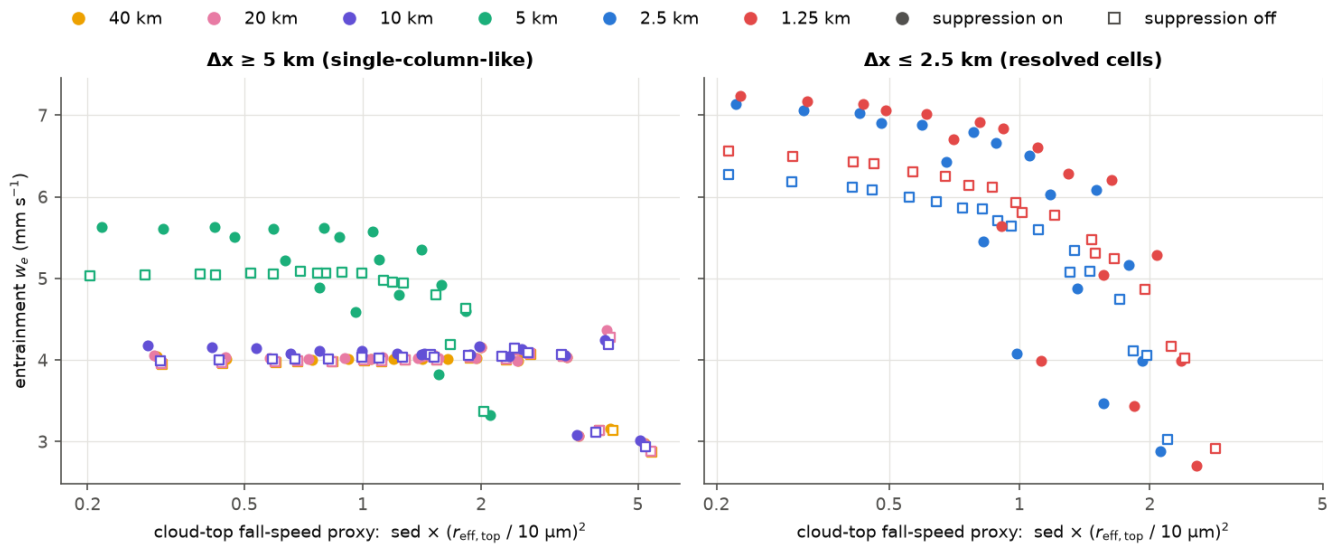


Figure 4: w_e against the cloud-top fall-speed proxy, hours 10–12. The $\geq$5 km panel now includes 5 km runs that have entered the resolved regime.

Data: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/` (runs) and `.../analysis/` (`summary_h10-12.csv`, `timeseries_15min.csv`, `susceptibility.csv`, `lwp_trend_h10-12.csv`, figures). **Sub-km (in progress):** 625 m and 312 m, 12 h, Nd 20–1000 × sed × suppression × 3D turbulence on/off (144 runs). They run in restart segments, which we checked are bit-for-bit, on interactive, debug and regular nodes.